

VALGA, Estonia

WMO: 262470

Lat: 57.7900N

Lon: 26.0378E

Elev: 66

StdP: 100.54

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.4	-18.3	-24.6	0.4	-22.2	-20.7	0.6	-17.7	6.7	-1.2	6.2	-1.1	0.8	60	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.0	19.5	26.3	18.5	24.6	17.8	20.8	26.0	19.7	24.4	18.7	23.2	2.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.9	23.2	17.9	13.0	22.1	16.9	12.1	20.9	60.3	26.2	56.5	24.4	53.1	23.1	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.6	5.0	DB	-25.0	30.5	4.6	1.7	-28.4	31.7	-31.1	32.7	-33.7	33.6	-37.0	34.8	(4)
(5)			WB	-25.3	22.4	4.5	1.3	-28.5	23.3	-31.2	24.1	-33.7	24.8	-37.0	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-4.8	-4.9	-0.7	5.9	11.2	15.2	18.0	16.5	11.7	6.1	1.3	-2.3	(6)	
	DBStd	9.21	6.39	6.63	4.57	4.18	4.29	3.33	3.09	2.94	3.33	4.02	4.53	5.64	(7)	
	HDD10.0	2173	460	417	332	134	35	2	0	0	21	131	261	381	(8)	
	HDD18.3	4504	718	650	591	372	223	103	44	73	199	381	511	639	(9)	
	CDD10.0	777	0	0	0	13	73	159	248	202	73	9	0	0	(10)	
	CDD18.3	64	0	0	0	0	3	11	34	16	1	0	0	0	(11)	
	CDH23.3	639	0	0	0	2	46	119	314	152	6	0	0	0	(12)	
	CDH26.7	117	0	0	0	0	6	18	69	24	0	0	0	0	(13)	
(14) Wind	WSAvg	2.3	2.5	2.5	2.5	2.4	2.1	2.1	1.9	1.9	2.0	2.3	2.5	2.7	(14)	
(15) Precipitation	PrecAvg	671	52	40	37	35	55	78	69	76	50	67	56	49	(15)	
	PrecMax	830	99	70	83	98	113	160	145	145	98	117	105	90	(16)	
	PrecMin	490	16	11	12	7	16	33	17	1	10	5	9	18	(17)	
	PrecStd	95	23	17	19	21	24	31	32	40	24	27	24	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	6.2	13.8	22.7	27.4	28.7	30.7	29.2	24.1	17.0	10.7	9.0	(19)	
		MCWB	6.4	4.5	7.0	13.1	18.2	19.4	21.3	20.1	16.8	13.3	9.1	8.0	(20)	
	2%	DB	4.6	4.6	9.6	19.4	24.4	26.6	28.6	26.8	21.0	14.4	9.1	6.9	(21)	
		MCWB	3.7	3.5	5.3	11.4	16.0	17.9	20.3	19.2	15.6	12.3	8.0	6.0	(22)	
	5%	DB	3.2	3.6	6.8	16.6	22.1	24.4	26.8	24.9	19.1	12.9	8.2	5.5	(23)	
		MCWB	2.5	2.6	3.8	9.8	14.3	17.4	19.4	18.1	14.8	11.4	7.3	4.8	(24)	
	10%	DB	2.1	2.6	5.0	14.0	19.7	22.2	24.9	23.0	17.4	11.7	7.1	4.1	(25)	
		MCWB	1.6	1.9	2.8	8.4	13.2	15.9	18.5	17.4	13.9	10.4	6.4	3.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	5.0	8.0	14.2	19.7	21.0	22.6	21.9	17.8	14.0	9.6	8.1	(27)	
		MCDB	6.8	5.6	12.9	21.1	25.6	26.5	28.8	27.0	21.3	15.6	10.3	8.9	(28)	
	2%	WB	3.9	3.7	5.7	12.2	17.3	19.4	21.3	20.1	16.5	12.7	8.3	6.1	(29)	
		MCDB	4.4	4.4	8.5	17.8	22.1	24.6	26.9	24.5	19.8	14.0	8.9	6.6	(30)	
	5%	WB	2.6	2.8	4.3	10.6	15.6	18.0	20.2	18.9	15.4	11.7	7.4	4.8	(31)	
		MCDB	3.0	3.4	6.3	15.4	20.1	22.7	25.0	23.4	18.2	12.9	8.0	5.5	(32)	
	10%	WB	1.6	1.9	3.2	9.0	14.0	16.8	19.2	17.9	14.4	10.6	6.4	3.5	(33)	
		MCDB	2.1	2.6	4.7	13.4	18.8	20.9	23.6	22.1	16.8	11.6	7.0	4.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.8	7.4	10.1	11.1	9.9	10.0	10.0	8.6	6.1	4.0	3.9	(35)	
		MCDBR	3.5	4.5	10.0	15.3	14.6	13.3	12.9	13.4	11.4	7.4	4.7	3.7	(36)	
	5% WB	MCWBR	3.3	3.8	6.5	8.0	7.2	6.2	5.7	6.3	6.5	5.4	4.1	3.6	(37)	
		MCWBR	3.5	4.0	8.2	13.2	12.1	11.1	11.5	11.4	9.7	6.8	4.3	3.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.277	0.307	0.338	0.374	0.368	0.368	0.394	0.390	0.359	0.336	0.317	0.272	(40)	
		taud	2.139	2.144	2.192	2.296	2.377	2.406	2.350	2.360	2.430	2.402	2.236	2.139	(41)	
	Edn at Noon		560	713	800	834	866	871	835	810	783	691	508	446	(42)	
		Edh at Noon	52	82	104	111	109	108	112	103	83	65	49	39	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.41	1.10	2.51	3.96	5.32	5.51	5.27	4.18	2.68	1.22	0.40	0.25	(44)	
	RadStd		0.09	0.20	0.33	0.46	0.55	0.43	0.51	0.41	0.30	0.18	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.43	N/A	N/A	N/A	N/A	N/A	-158	N/A	N/A	N/A	(46)
(47)	Regional (34 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	-142	N/A	N/A	N/A	(47)

Nomenclature: See separate page